Final title I'd lock

Spatial Multimodal Edge-AI for Fault Detection and Localization of Rotating Machinery

That is broad enough that the **major can scale to industrial machinery**, while your minor can safely prove the concept on a small DC motor.

The distinction is:

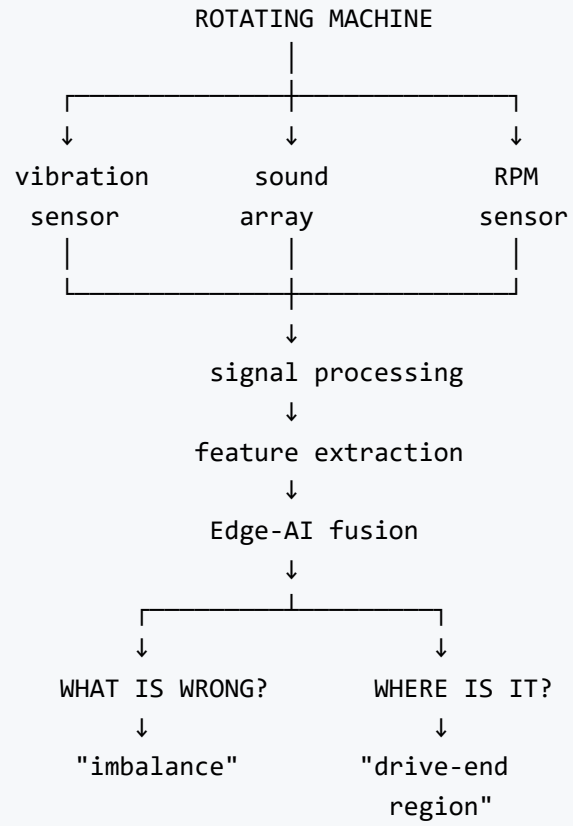
```
PROJECT SCOPE:
Rotating machinery
        ↓
MINOR PROTOTYPE:
small safe DC motor
        ↓
MAJOR:
larger / industrial machinery using properly rated,
isolated sensing hardware
```

You do **not** need to electrically connect an ESP32 directly to a high-voltage machine for the project to work on high-voltage machinery. The sensors and acquisition front end are what interface to the machine, with appropriate isolation and ratings.

Now forget all the buzzwords for a second

What are we actually building?

Imagine this:



So the machine isn't just saying:

> FAULT DETECTED

It's trying to answer:

> What is happening, where is it happening, and how confident are we?

That spatial part is where the acoustic array becomes important. Acoustic beamforming is specifically used for sound-source localization and has applications in mechanical fault diagnosis. (ScienceDirect)

What each sensor actually does

1. Vibration sensor

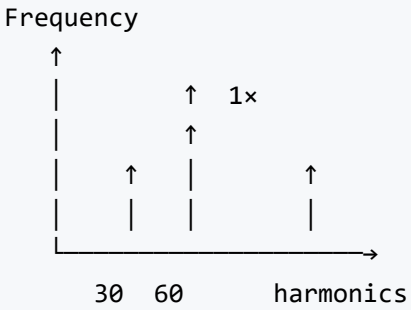
An accelerometer is attached to the machine.

Suppose the shaft is rotating at 1800 RPM:

$$f_r = \frac{1800}{60} = 30 \text{ Hz}$$

A mechanical problem can change the vibration spectrum.

You might see:



The vibration channel tells us a lot about **mechanical condition**.

---

2. Acoustic array

Instead of one microphone:

🎤

we use several:

🎤 🎤
🎤
🎤 🎤

A sound generated at one point on the machine reaches each microphone at slightly different times.

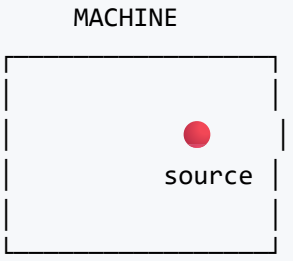
For example:

```
mic 1: 100.000 ms
mic 2: 100.037 ms
mic 3: 100.061 ms
mic 4: 100.084 ms
```

Those tiny differences contain spatial information.

We calculate things like **TDOA**, time difference of arrival, and use beamforming/SRP-PHAT-type processing to estimate where the acoustic source is. Beamforming is essentially spatial filtering by mathematically steering an array toward possible source locations. (ScienceDirect)

The output becomes:



This is why your earlier heatmap matters.

---

3. RPM sensor

RPM tells us what the machine is actually doing.

Suppose:

$$RPM = 1800$$

Then:

$$1X = 30\ \text{Hz}$$

Now suppose the vibration spectrum suddenly has a large peak at 30 Hz.

That's much more informative than simply saying:

> "30 Hz is strong."

We know it's **1× shaft rotational frequency**.

That lets us use **order analysis**, which becomes important as the project gets more advanced.

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4. AI

The AI isn't supposed to magically stare at raw sensor data and announce:

> "bearing broken."

Instead, we can give it meaningful measurements:

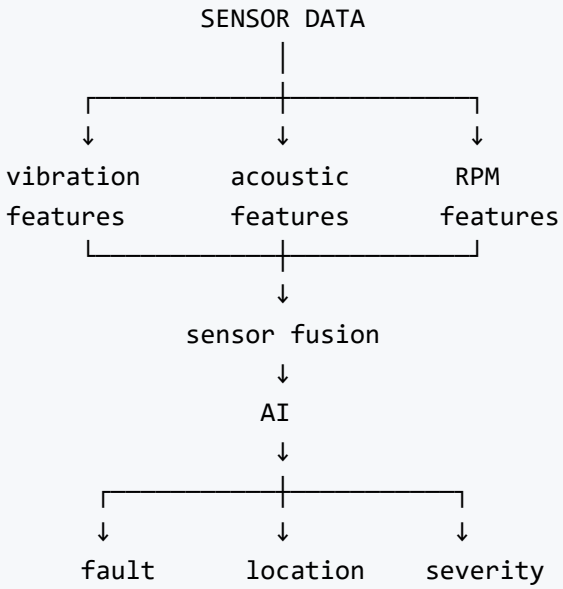
```
Vibration:
RMS
peak
kurtosis
spectral energy
1× amplitude
2× amplitude
harmonics

Acoustic:
source position
acoustic energy
spectral features
impulsiveness
beamformed energy

RPM:
speed
order information
```

Then the model learns relationships between them.

Something like:



What does "multimodal" actually mean?

It simply means:

> **Different types of measurements describe the same physical event.**

For example:

Normal

```
vibration = normal
acoustic = normal
RPM = stable

→ HEALTHY
```

Imbalance

```
vibration ↑ strongly at 1×
acoustic source near rotor
RPM stable

→ POSSIBLE IMBALANCE
```

Something near a bearing

```
high-frequency vibration
+
impulsive acoustic energy
+
acoustic hotspot near bearing region

→ POSSIBLE BEARING-RELATED EVENT
```

That's sensor fusion.

What makes this different from the GitHub repo?

The GitHub project is mainly doing:

```
vibration
   ↓
FFT/features
   ↓
ML
   ↓
fault classification
```

Its documented system uses MPU6050 vibration, current/temperature sensing, FFT-derived features and an INT8 TFLite model on a Raspberry Pi. (ScienceDirect)

Our architecture is:

```
vibration ──────┐
                │
acoustic array ├──→ spatial + multimodal fusion
                │
RPM ────────────┘
                      ↓
                   Edge-AI
                      ↓
              WHAT + WHERE
```

That's a **real architectural difference**, not just swapping an Arduino board.

What the minor should actually prove

Don't try to build the entire final machine in semester one.

The minor should answer:

Can we detect a controlled mechanical fault and spatially localize the dominant acoustic source?

Use a small motor.

For example:

```
        MOTOR
      ┌────────┐
      │        │
======| shaft  |======
      └────────┘
       ↑      ↑
    sensor  bearing
```

And test controlled conditions:

```
NORMAL
IMBALANCE
MISALIGNMENT
LOOSENESS
```

Then:

```
accelerometer → mechanical signature
microphone array → acoustic location
RPM → operating condition
             ↓
        laptop fusion
             ↓
   fault + spatial heatmap
```

That is enough for a very strong minor.

Then the major becomes the real edge system

The minor:

```
Sensors
   ↓
Laptop
   ↓
DSP + AI
```

The major:

```
Sensors
   ↓
custom acquisition hardware
   ↓
MCU / edge processor
   ↓
real-time DSP
   ↓
Edge-AI
   ↓
standalone diagnostic instrument
```

The ESP32-S3 is technically capable of the kind of multichannel digital-audio acquisition we're discussing. Its I2S hardware has DMA and supports TDM, and Espressif documents a four-microphone TDM recording example. (Espressif Documentation)

So **ESP32-S3 is a reasonable prototype target**, but we don't need to commit the entire project to it before we know the sensor architecture works.

The biggest future difficulties

This is the part I really want you to understand before committing.

1. Acoustic reflections

Risk: VERY HIGH

This is probably the hardest part of the acoustic side.

Your lab isn't:

```
perfect empty room
```

It's:

```
machine
 ↓
wall
 ↓
table
 ↓
floor
 ↓
other machines
 ↓
reflections
```

The microphone array can receive:

- direct sound
- reflected sound
- unrelated background noise

Beamforming literature treats array calibration, interference and correction as major practical issues. (ScienceDirect)

What we do

Start in a controlled setup.

Then deliberately add noise/reflections and measure degradation.

That itself becomes useful experimental work.

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2. Microphone geometry

Risk: HIGH

Your current laptop test uses channels, but we don't yet know the exact physical coordinates of its microphones.

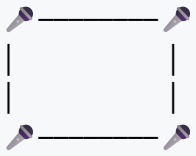
Beamforming depends on knowing:

$$(x_i, y_i, z_i)$$

for each microphone.

So our physical prototype should eventually use a known geometry.

Something like:



Measure the spacing precisely.

---

3. Synchronization

Risk: HIGH later

The array only works if channels are properly synchronized.

A tiny timing error becomes an angle error.

That's why you can't just have:

```
4 completely independent cheap recorders
```

and casually combine their audio.

You want a shared clock / synchronized acquisition architecture.

That's exactly why the eventual hardware design matters.

---

4. Small array vs spatial resolution

Risk: MEDIUM/HIGH

You cannot have:



spread over 3 cm and expect the resolution of a professional industrial acoustic camera.

Array aperture, frequency and wavelength matter.

This is a fundamental tradeoff in acoustic imaging and array design. (ScienceDirect)

So we must select:

- number of microphones
- spacing
- frequency band
- distance to machine

together.

---

5. Real fault data

Risk: VERY HIGH

This is actually the biggest **AI** problem.

It's easy to generate:

```
normal = 100 recordings
imbalance = 100 recordings
```

But it is much harder to obtain meaningful realistic bearing degradation data.

So:

Minor

Use reproducible faults we can safely create.

Major

Add real datasets and/or better controlled degradation experiments.

Don't pretend we have industrial-grade bearing failure data when we don't.

---

6. Acoustic fault != mechanical fault

This is extremely important.

A microphone hears:

```
motor
fan
table
person
air conditioner
keyboard
other machine
```

So an acoustic hotspot does **not automatically mean fault**.

That's why we need multimodal sensing.

For example:

```
ACOUSTIC:
strong

VIBRATION:
normal

→ probably not enough evidence for mechanical fault
```

But:

```
ACOUSTIC:
strong + localized

VIBRATION:
abnormal

RPM:
correlated frequency change

→ much stronger evidence
```

That's one of the main reasons this multimodal project makes sense.

---

7. AI overfitting

Risk: HIGH

If we train on:

```
Motor A
Lab A
RPM = 1800
```

then test:

```
Motor A
Lab A
RPM = 1800
```

and get:

99% accuracy

that doesn't mean much.

The model may simply memorize the environment.

So eventually we need tests across:

- different RPM
- different loads
- different positions
- different motor runs
- background noise
- possibly different motors

That becomes a major-project research challenge.

---

8. Edge computing

Risk: MEDIUM initially, HIGH later

Laptop:

```
lots of RAM
CPU/GPU
Python
```

is easy.

ESP32:

```
limited RAM
limited compute
real-time deadlines
```

is not.

But that's exactly why the project has a nice minor → major progression.

One thing I would NOT do

I would **not** start with:

```
"Let's train an AI model."
```

Start with physics.

First:

```
motor
 ↓
raw signals
 ↓
FFT
 ↓
RPM relationship
```

Then:

```
microphones
 ↓
TDOA
 ↓
beamforming
 ↓
heatmap
```

Then:

```
vibration + acoustic + RPM
 ↓
feature fusion
```

Then:

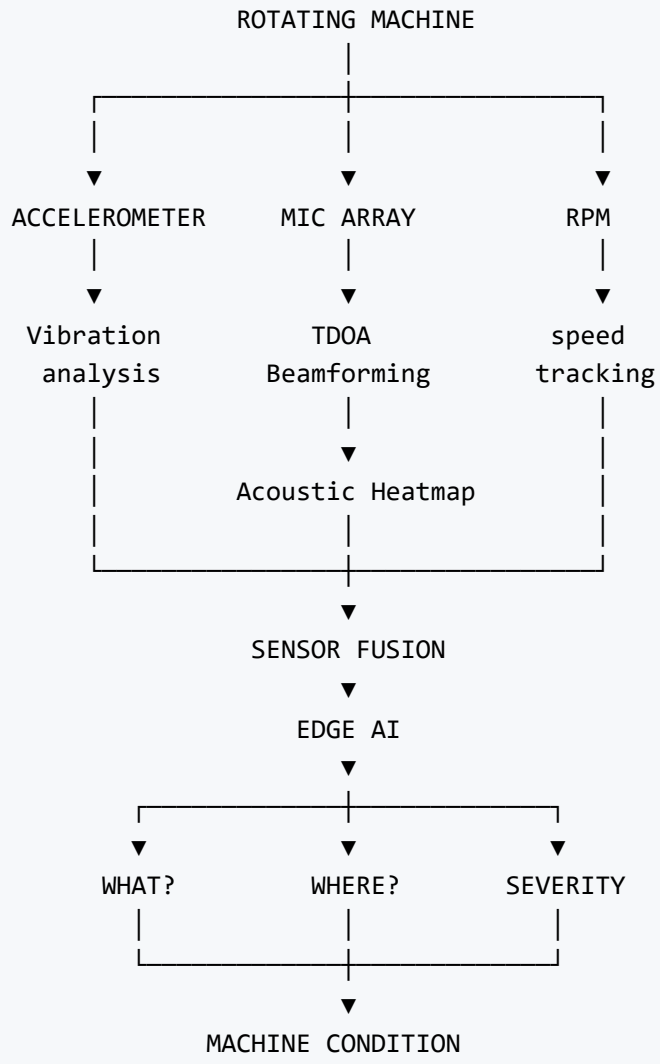
```
AI
```

Then:

```
edge deployment
```

That's the order.

So the whole project in one picture



And **eventually**, the major can add:

```
condition history
       ↓
degradation trend
       ↓
early-warning / prediction
```

That's where the predictive-maintenance part comes in.

One final naming point

I would **not put "predictive maintenance" in the title yet**.

Use:

> Spatial Multimodal Edge-AI for Fault Detection and Localization of Rotating Machinery

Then your major can legitimately add:

> ...with Degradation Tracking and Predictive Maintenance

once you've actually built the longitudinal degradation part.

That keeps the title technically honest and gives you room to grow instead of promising a capability before you've got the data for it.

And yes, **"Rotating Machinery" without the low-voltage qualifier is the right wording**. The prototype can be a safe low-voltage motor, while the engineering architecture is intended to generalize to larger machinery through appropriate sensor interfaces and isolation. The acoustic localization portion is also grounded in established mechanical fault-diagnosis research, rather than being an invented application. (ScienceDirect)